ISRAT JAHAN LAMIA

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Education

BRAC University — September 2026

Bachelor of Science in Computer Science and Engineering (CSE) — *Dhaka, Bangladesh*

- **CGPA:** 3.3/4.0
- **Major:** Computer Science and Engineering

Research Interests

- Artificial Intelligence (AI) and Machine Learning
- Deep Learning and Computer Vision
- Supervised, Unsupervised, and Reinforcement Learning
- Ensemble Learning, Classification, and Regression Modeling
- Feature Engineering and Predictive Analytics
- Medical Image Processing and Analysis
- Biomedical Engineering
- Computational Modeling and Simulation

Research Experience

Publications — 2025

Dielectrophoresis-Based Cell Separation and Controlled Transportation in Microfluidics System (Paper)

Unveiling the Optimal CNN-A Performance Evaluation of Architectures in Malaria Image Classification (Paper)

Stacking Ensemble of Machine Learning Methods for Brain Tumor Size Prediction (Accepted for Publication)

Other Papers | *Researched*

- Conducted research in Artificial Intelligence, Machine Learning, and Biomedical Engineering applications.
- Worked with deep learning models for malaria image classification and performance comparison of CNN architectures.
- Applied machine learning and ensemble learning techniques for biomedical prediction tasks.
- Participated in the development and evaluation of predictive models for healthcare-related applications.
- Designed and simulated a dielectrophoresis (DEP)-based microfluidic system using COMSOL Multiphysics.
- Performed computational modeling and simulation for biomedical engineering applications.
- Contributed to the preparation and publication of peer-reviewed research papers.

BRAC University | *Undergraduate Thesis* — 2025

- Brain Hemorrhage Detection after Brain Stroke using Deep Learning Algorithms
- Conducted literature review on brain hemorrhage detection using CT and MRI images.
- Explored deep learning techniques for medical image classification and automated diagnosis.
- Analyzed challenges in post-stroke hemorrhage detection and AI-assisted clinical decision support systems.

Work Experience

Codium Digital May 2026 – Present
Software Engineer Intern *Remote, Bangladesh*

- Worked as a Software Engineer, contributing to the design, development, and maintenance of software applications.
- Developed scalable backend and frontend features, collaborated with cross-functional teams, and improved application performance.
- Participated in code reviews, debugging, deployment, and implementation of production-ready software solutions.

Relevant Course

- ARTIFICIAL INTELLIGENCE
- VLSI DESIGN
- INTRODUCTION TO ROBOTICS
- DATA STRUCTURES
- DATABASE SYSTEMS
- MACHINE LEARNING
- ALGORITHMS
- COMPUTER ARCHITECTURE
- OPERATING SYSTEMS
- DATA COMMUNICATIONS

Academic Projects

Visual Fashion Stylish – AI Inspired Personalized Fashion Styling Platform [GitHub]

- Developed a fashion recommendation platform using HTML, CSS,JavaScript and PHP with user authentication, profile management, and personalized dashboards.
- Implemented skin tone detection and body shape analysis to provide customized color suggestions and outfit recommendations.
- Built virtual wardrobe, favorites management, and e-commerce integration features for personalized styling and seamless shopping experiences.

Disaster Relief Coordination System (Ongoing) — MERN Stack [GitHub]

- Developing a full-stack disaster management platform using MongoDB, Express.js, React.js, and Node.js with JWT-based authentication and role-based access control.
- Implementing multi-role dashboards for citizens, volunteers, and administrators with emergency request tracking, shelter management, and rescue task assignment.
- Integrating Google Maps API, Cloudinary, Nodemailer, and Twilio APIs for location services, image uploads, email notifications, and SMS alerts.

Bracu Diaries [GitHub]

- User authentication with role-based dashboard access.
- Interactive diary/blog writing with Markdown support.
- Secure file upload for past question papers and notes.

IoT-Based Smart Rickshaw Safety System — Arduino, C++, GPS, GSM, IoT [GitHub]

- Built an Arduino-based smart transportation system with GPS, GSM, ultrasonic sensor, RTC, and LCD integration.
- Implemented overspeed detection, illegal parking alerts, and one-touch SOS notifications with real-time location tracking.
- Designed a low-cost IoT solution for intelligent traffic monitoring and passenger safety using embedded systems.

Technical Skills

Programming Languages: Python, HTML, CSS, C, JS

Frontend: HTML, CSS, React.js

Backend: Node.js, Express.js

Database: MongoDB, MySQL

Development Tools: GitHub, VS Code, phpMyAdmin

Simulation Software: Google Colab, Matlab

Hardware: Arduino, Raspberry Pi

Leadership & Extracurricular Activities

- Member, IEEE BRAC University Student Branch.
- Participated in the GHEA21 Virtual Student Leadership Conference 2025, focusing on leadership, teamwork, and professional development.
- Participated in the Webinar on Artificial Intelligence organized by Bangladesh Foundry & Engineering Works Limited (BFEW) Technical Training Center.
- Completed the *Machine Learning A-Z: AI, Python & R + ChatGPT Prize [2025]* course (43 hours) on Udemy.

Achievements and Awards

PSC board scholarship for outstanding results

JSC board scholarship for outstanding results

Reference

Salaudin Gazi
Android Technical Lead at **meldCX**
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